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Multi-metric Analysis of the Relation Between
Movement Complexity and Creativity
Measures
Master's Thesis

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Multi-metric Analysis of the Relation Between Movement Complexity and Creativity Measures

Abstract:

Creativity is commonly assessed using behavioral tasks that measure different aspects of creative cognition. Recent theory has suggested that creativity may reflect exploratory search processes within conceptual space, and prior research has linked movement complexity with creative thinking. However, experimental evidence of the connection between exploratory movement complexity and creativity is unclear. This study investigates whether exploratory movement behavior in virtual reality (VR) environments is associated with creativity measures. Movement trajectories were collected while participants performed a virtual foraging task, and their creativity was assessed using the Alternative Uses Test (AUT), Divergent Association Task (DAT), and Creative Self-Efficacy. To quantify exploratory behavior, 18 movement metrics spanning spatial, entropic, and kinematic analyses were calculated and compared against creativity measures. Different dimensions of creativity were associated with different forms of movement complexity, with Divergent Association Task scores aligned more strongly with micro-adjustments, while Alternative Uses Test scores were associated with broader exploratory movement patterns. The trajectory metrics capture different properties of exploratory movement behavior, showcasing the potential for multi-metric approaches for future movement analysis and creativity research.

Keywords: movement complexity, trajectory analysis, feature extraction, virtual reality, creativity assessment, cognitive foraging

CERCS:

P170 Computer science, numerical analysis, systems, control

S260 Psychology

Liikumise keerukuse ja loovuse mõõdikute vahelise seose mitmemõõtmeline analüüs

Lühikokkuvõte:

Loovust hinnatakse erinevate käitumuslike ülesannete abil, mis mõõdavad selle eri aspekte. Hiljuti välja pakutud teooria kohaselt võib loovust mõista kui otsimiskäitumist vaimses ruumis. Kas see otsimiskäitumine väljendub liikumisena füüsilises ruumis on aga ebaselge. Käesoleva töö eesmärgiks on uurida, kas liikumismustrid otsimiskäitumise ajal virtuaalreaalsuses (VR) on seotud loovuse mõõdikutega. Liikumismustrite kvantifitseerimiseks kasutati 18 erinevat mõõdikut, mille hulgas olid ruumilised, entroopilised ning kinemaatilised analüüsid. Liikumistrajektoore mõõdeti virtuaalreaalsuse metsas toidu otsimise ajal. Loovust hinnati kasutades Alternatiivsete kasutusviiside testi (AUT), Divergentsete Assotsiatsioonide Testi (DAT) ja enesehinnangulist loova enesetõhususe küsimustikku. Liikumistrajektooride ja loovuse mõõdikuid omavahel võrreldes selgus, et erinevad trajektoori mõõdikud seostusid erinevate loovuse mõõdikutega. DAT tulemused seostusid rohkem mikromuutustega trajektooris ja AUT tulemused rohkem suurema skaala liikumismustritega. Kõik trajektoormõõdikud hindavad liikumiskäitumise erinevaid aspekte. See tõstab esile multimeetrilise lähenemisviisi eelise tulevasteks uuringutes liikumiskäitumise ja loovuse valdkonnas.

Võtmesõnad: liikumise keerukus, trajektoori analüüs, tunnuste eraldamine, virtuaalreaalsus, loovuse hindamine, kognitiivne toitumise otsimine

CERCS:

P170 Arvutiteadus, arvutusmeetodid, süsteemid, juhtimine (automaatjuhtimisteooria)

S260 Psühholoogia

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1. Introduction

Creativity is typically assessed through a variety of different tasks, each of which measures different aspects of cognitive creativity. Traditional approaches have largely focused on divergent thinking by evaluating the responses generated during structured tasks [1, 2]. However, recent studies have suggested that creativity may reflect exploratory search processes within conceptual space. This draws parallels between exploration within conceptual and physical space, likened to foraging for physical and conceptual resources [3, 4]. As such, exploratory behavior may contain measurable signals related to creativity.

Previous research has linked movement complexity with creativity, particularly through hand complexity and exploratory behavior [5, 6]. Virtual Reality (VR) provides an opportunity to expand on these approaches by enabling detailed tracking methods within controlled environments [7, 8]. Current advances in data tracking make it possible to use different metrics to quantify the complexity of trajectories, but there is no well established method for determining which characteristics are most relevant to creativity-related behavior. This raises several questions regarding whether measurable movement patterns are correlated with creativity, and which are more informative to describe movement complexity from exploratory behavior. Whether different creativity measures correspond to different movement characteristics is also unclear.

In this study, VR was used to collect movement data while subjects searched for berries in a virtual forest. Multiple creativity assessments were used to capture various aspects of creativity, including the Alternate Uses Test (AUT), Divergent Association Task (DAT), and Beghetto's Creative Self-Efficacy [9–11]. Movement captured from the subjects was measured using 18 complexity metrics. Dimensionality reduction and comparative analyses were then performed to explore potential relationships between movement complexity metrics and creativity measures.

Given the absence of established frameworks linking exploratory movement trajectories to creativity measures, this study adopts an exploratory analytical approach. Rather than relying on a predefined movement metric, a range of theoretically motivated trajectory measures were evaluated to investigate potential behavioral relationships and characterize the multidimensional structure of exploratory movement complexity.

This work introduces a multi-metric pipeline for quantifying movement complexity across multiple behavioral dimensions using data collected from a VR-based framework. By combining

creativity assessments with computational trajectory analysis, this study explores whether measurable patterns in movement behavior are associated with aspects of creative cognition.

Generative AI tools were used for minor language editing and formatting assistance during manuscript preparation. The study design, analysis, interpretation and scientific content were developed by the authors.

2. Background

2.1 Creativity and Cognitive Measures

Creativity is a highly complex phenomenon which humans have tried to study for hundreds of years. However, despite the endless research, psychology still lacks a scientific consensus [1, 12]. But in the 20th century, a significant effort was made to define creativity [13], in which early definitions described divergent thinking as a key component of creative cognition [14]. While modern ideas have incorporated convergent thinking into the definition, divergent thinking remains the most widely used basis for measurement [1, 2].

Divergent thinking is typically measured through structured experiments designed to assess the ability to generate novel ideas. These tasks often evaluate components such as fluency, originality and flexibility, providing a quantifiable approximation of creative cognitive ability [1, 15]. One of the most widely used measures for creativity is the Alternative Uses Test (AUT) due to its simplicity, flexibility and long history of evaluation. In the AUT, which was originally introduced by Guilford, participants are asked to generate unconventional uses for common objects [9].

Despite the AUT’s extensive usage, the test is not without limitations. Notably, scoring is subjective as responses must be evaluated by human raters. This introduces bias between raters and can reduce the reliability or consistency of results across studies [2, 16]. As a result, recent studies have developed objective and automated methods for assessing creativity [10].

The desire for an objective approach led to the rise of a new scoring method for divergent thinking called the Divergent Association Task (DAT). Unlike the AUT, which relies on subjective human evaluation, the DAT provides a fully automated and quantitative measure. In this task, participants are asked to generate a set of unrelated words, and creativity is assessed based on the semantic distance between those words [10]. Semantic distance between responses has been shown to reflect the ability to connect remote concepts within semantic memory, indicating creative thinking [17].

To compute semantic distance, words are first represented as vectors in a high-dimensional space. These vector models are based on the idea that words appearing in similar contexts have similar meanings, allowing semantic relationships to be numerically quantified [18]. One of the foundational methods for generating such representations is Word2Vec, which learns word embeddings from a large text corpus by modeling the contexts where words appear [18].

However, Word2Vec focuses on whole words, which can limit its effectiveness in languages with rich morphology or large vocabularies. To address this issue, FastText expands upon the approach by representing words as combinations of character n-grams, therefore incorporating sub-word information [19]. This is particularly useful in agglutinative languages where sub-words and suffixes carry significant meaning such as Finnish and Estonian. The ability to separate sub-words as independent tokens allows greater handling of morphological structure and rare words [20, 21].

Once word embeddings are obtained, similarity metrics such as cosine similarity can be used to calculate the semantic distance between words. Cosine similarity measures the angle between two vectors, providing a normalized measure of their semantic relatedness. This approach has been widely used in natural language processing and has been shown to be effective in capturing meaningful relationships between words [22]. Within the DAT test, lower similarity between words indicates greater semantic distance and corresponds to higher creativity scores. With this simple scale, the DAT provides an objective and scalable alternative to traditional divergent thinking assessments [10].

However, modern definitions of creativity increasingly include convergent thinking as another key component of creative cognition [23]. One widely used measure of convergent thinking is the Remote Associates test (RAT), which requires participants to identify a common link between seemingly unrelated words [24]. Unlike divergent thinking tasks where subjects generate responses, the RAT is highly dependent on language specific inputs of standardized word sets [16, 25]. This dependence makes similar tests difficult to use reliably in uncommon or morphologically rich languages such as Estonian, where validated and standardized test materials are not readily available [22, 25].

In addition to these tests, self-reporting creativity measures can provide additional information that could be missed such as attitudes and creative identity. Inherently, self-reporting measures are influenced by self-awareness and personal bias[26, 27]. One such method for self-reporting is Beghetto's Creative Self-Efficacy, which involves asking participants to rate their own creativity in a standardized questionnaire [11]. In addition to creative self-perception, cognitive style measures assess preferred ways of thinking using several cognitive dimensions, such as knowing, planning, and creating. These dimensions are not direct measures of creativity, but imply relations to creative behavior. Creating is associated with creative tendencies and planning may

show opposing relationships [28]. Including these measures allows for a comparison between a subject's internal perceptions and performance-based assessments.

While these tests provide powerful tools for measuring creativity, they do not directly capture the mechanisms and processes that give rise to creative thinking [1]. As a result of these limitations, various alternative approaches have been developed to capture creativity as a dynamic process rather than a static outcome.

The theory that creativity can be conceptualized as a search process has been developed, drawing parallels between cognitive processes and spatial foraging behaviors. This indicates that generating new ideas can be reflective of exploring a conceptual space, similar to how humans or animals might search for resources in a physical environment. This suggests that creative thinking may align with underlying mechanisms shared with exploratory behavior [3, 4].

One proposed basis for this link is the SEEKING system, a dopamine-based network associated with motivation and exploration [29]. Dopamine has been shown to play a significant role in the seeking of new ideas and exploratory behavior, particularly in animal studies [29–31]. While not directly equated with creativity, this supports the idea that creative processes and cognitive exploration may share common behavioral mechanisms [3, 4].

Unlike many cognitive processes, exploratory behavior can be physically observed through movement. Studies have shown that more complex movement trajectories are associated with higher creativity. For example, Peng and Braun demonstrated that the complexity of hand movement trajectories correlated with subjective creativity rankings [5]. Similarly, other work has shown that movement patterns become more complex when individuals are instructed to behave creatively [3, 6]. These findings suggest that movement trajectories may provide a measurable proxy for exploratory and creative processes [3, 5].

To capture movement data with high precision, virtual reality (VR) environments can be used as they provide a controllable platform while allowing for continuous tracking of head and hand movement [7, 8]. However, VR has also been shown to influence cognitive processes such as creativity. Prior work has suggested that VR supports creative thinking and potentially alters behavior [32]. Due to the effects of VR on creativity, participants' prior VR experience was recorded as a potential confounding factor in this study.

2.2 Trajectory Analysis

To analyze movement trajectories, a wide range of methods to quantify movement complexity are available. One of the earliest and most influential measures still in use is Shannon entropy, which was created in 1948 by Shannon. Shannon entropy quantifies the uncertainty of a system and can be applied to movement data to measure the predictability of the trajectory. When spatial positions are more dispersed, the distribution becomes less predictable and trajectories exhibit high entropy [33]. However, Shannon entropy does not use temporal data, focusing only on positional distribution. To measure entropy over temporal data, measures such as Markov trajectory entropy incorporate the probabilities between different states, enabling an analysis of sequence-dependent entropy [34].

While entropy provides a measure of uncertainty in movement, its application depends on how spatial data is represented. A common procedure in symbolic dynamics is to discretize trajectories into symbolic representations to facilitate the use of entropic measures [35]. One example is grid-based representations such as occupancy maps and heatmaps that provide a valuable discretization that allows for further measurement [36, 37].

These representations provide tools to analyze spatial concentration and distribution within trajectories [35]. For example, heatmaps can be used to identify regions of high and low occupancy, providing insight into how movement is distributed across the environment [37]. These analyses allow researchers to distinguish between various movement patterns, which may reflect different exploratory behaviors [3, 28, 31].

In addition to entropic and spatial distribution methods, continuous trajectory based metrics supply further insight into movement dynamics. One widely used measure is Mean Squared Displacement (MSD), which quantifies the average distance traveled over time and is commonly used to identify patterns such as random, constrained, or directed motion [38]. By analyzing how displacement evolves across time intervals, MSD captures temporal aspects of movement that are not reflected in purely spatial measures.

Another important metric is tortuosity, which describes the degree of curvature or deviation in a movement path. Highly tortuous trajectories indicate frequent changes in direction, while low tortuosity reflects more direct movement. This measure has been widely applied in studies of animal movement and navigation to distinguish between different exploration strategies [39].

Together, these trajectory-based metrics complement entropy and spatial analyses by capturing how movement unfolds over time, resulting in a more comprehensive view of exploratory behavior.

Given the large variety of trajectory metrics, dimensionality reduction techniques can contribute a powerful tool for investigating hidden structures in the data. One commonly used method is Principal Component Analysis (PCA), which transforms a set of potentially correlated variables into a smaller number of orthogonal components to capture as much variance in the data as possible [40]. By projecting high-dimensional feature spaces into lower-dimensional representations, PCA helps identify dominant patterns while reducing redundant features.

PCA has been used for movement analysis to summarize complex behavioral data and reveal latent structures [41]. This allows measures such as entropy, trajectory kinematics, and spatial distribution to be analyzed jointly instead of in isolation. As a result, PCA is a useful tool for comparing movement patterns and identifying relationships between behavioral features.

These approaches establish a foundation for analyzing both creativity and movement complexity using quantitative methods that can be used for computation and comparison. Creativity can be assessed through a combination of behavioral tasks and computational metrics, while movement trajectories can be characterized through spatial, temporal, and statistical analyses. These complementary perspectives provide the basis for examining potential relationships between exploratory behavior and creative processes.

3. Methods

3.1 Creative Test Design and Virtual Environment

The experimental process was divided into two primary stages: the creative experiment and the trajectory analysis. Two primary creativity tests were performed, the Alternative Uses Test (AUT) and the Divergent Association Task (DAT). In addition, self-reported creativity and a cognitive styles test were performed. A more detailed analysis and motivation for these cognitive measures are presented in the complementary thesis by Kaljo (2026), which emphasizes the psychological interpretation of the experimental framework and creativity measures.

Both the AUT and DAT measure divergent thinking, a key aspect of creativity. AUT measures divergent thinking by providing the subjects with an object to focus on, in this case, a paper clip. The participants are then given three minutes to come up with as many novel applications for the object as possible, identifying their flexibility and originality. These tests were recorded and later required manual transcription. With these transcriptions, answers were encoded into a reduced form which allowed the final scores to be counted and calculated into four different scores. Further details of the transcription and encoding methodology can be found in Kaljo (2026).

The first score is AUT Fluency, which is a metric for how many ideas the subject had throughout the test. AUT Elaboration then measures how many thoughts per named idea that each subject had, therefore measuring how much they 'elaborated'. AUT originality compared each of the ideas to each other, including the ideas from all studies. This was then measured to see how common their ideas were, identifying how original the subject's thought process was. Lastly was AUT Flexibility, which is the most bias-prone test of the four. Various categories were created, separated all ideas across all subjects into separate bins. AUT Flexibility measured how many categories each participant developed ideas in. Given that cosine distance had already been utilized for this research, it was used again to measure the average cosine distance between the reduced form of the subject's answers, creating the AUT cos metric.

In contrast, DAT quantifies the ability for the subject to navigate their associative memory and gauge the semantic distance between unrelated concepts. Subjects were instructed to produce ten words that were as semantically distant as possible within one minute. In order to compare the semantics of the words used, word embeddings were retrieved from fastText. fastText was used over Word2Vec due to the complex morphological structure of Estonian, which fastText's models are better for due to the ability to measure subwords. fastText uses a Continuous Bag

of Words (CBOW) with position weights on Wikipedia and Common Crawl. These use 300 dimensions, with a window of size 5, n-grams of length 5, and 10 negatives. fastText provides these pretrained models for 157 different languages, including Estonian, which is the language used in this experiment.

Semantic distance was calculated as the mean pairwise cosine distance between word vectors, which were retrieved for each word from fastText's pretrained model. However, some subjects did not properly complete the task, only writing down eight or nine words instead of the required ten. To maintain reliability within the data, three different categories were created for the final DAT measurements. DAT_8 contains the score from the first eight words that each subject submitted. DAT_9 contains the scores from the first nine words that each subject submitted, but for those that only submitted eight, their DAT_8 score was used. DAT_cos_max included the scores of all ten words where possible and scores from DAT_8 or DAT_9 where necessary.

CP_mean created the average of three scores provided by a small survey, where the subjects reported their perspective on their own creativity. Afterwards, a longer survey of 34 questions was created to measure the various dimensions of cognitive styles which are not mutually exclusive. These styles included 'knowing', 'planning', and 'creating' and were written as K_CS_mean, P_CS_mean, and C_CS_mean accordingly. The metrics were created by calculating the mean score of the questions assigned to each dimension. Notably, the survey design accounts for the fluid nature of cognition by allowing specific items to contribute to multiple dimensions, reflecting the overlap between them. This multi-dimensional profile allows for a more nuanced correlation analysis between a subject's self-reported cognitive habits and their physical movement complexity in the VR environment.

To observe the behavioral process of the subjects, a virtual reality environment was designed by Madli Kaljo and Jared Michael Wildermuth. This area was four by six square meters, with one corner having a square meter removed due to a desk occupying the space.

The subjects began with a 'tutorial environment', which was created as a separate application. In the tutorial, a simple grey platform was provided to stand on, with several clearly visible berries. A basket was also provided for the subjects to drop the berries in, or the basket itself could be used to scoop up the berries. The inside of the basket contained collision which trapped the berries, preventing them from being dropped, a common issue with initial testing. Once users were comfortable with the system, they could move on to the tracked experiment.

The backdrop of the virtual environment was that of a foggy forest, with a tall ring of bushes marking the boundaries of the room as can be seen in Figure 1. Many participants pushed a small amount into the bushes, but the combination between the open space surrounding the area and the occasional guidance from the experimenter avoided collisions. A large rock was used to block off the desk, as it provided a much harder boundary than the bushes. The floor was covered in grass, hiding seven berries within the environment, which were placed in randomized locations.



Figure 1. An image of the VR environment, with the bushes being the main focus. The background can be seen, as well as some of the ground and flowers.

Participants had an unlimited amount of time to view the environment itself until they picked up the basket, which generated the berries and started a timer. Upon the timer starting, subjects had three minutes to find as many berries as possible. This time constraint was enforced to ensure that every participant remained in the experiment for the same amount of time, while providing a long enough trajectory to run tests. Subjects were informed that there were more berries than actually existed to ensure they kept searching in case all berries were found.

3.2 Feature Extraction for Trajectory Data

3.2.1 Data Description

With the Meta Quest 3's internal system, the participant's head position was continuously tracked and recorded across the X, Y, and Z axis. The data was sampled at a frequency of 10 Hz, resulting

in a total of 1800 data points over three minutes. This sampling rate was selected to track both macro and micro-level navigation. These coordinates were extracted from the Unreal Engine log files and compiled into a movement trajectory as seen by Figure 2. This data was then stored and attached to a randomized ID-code to provide an additional layer of anonymity.

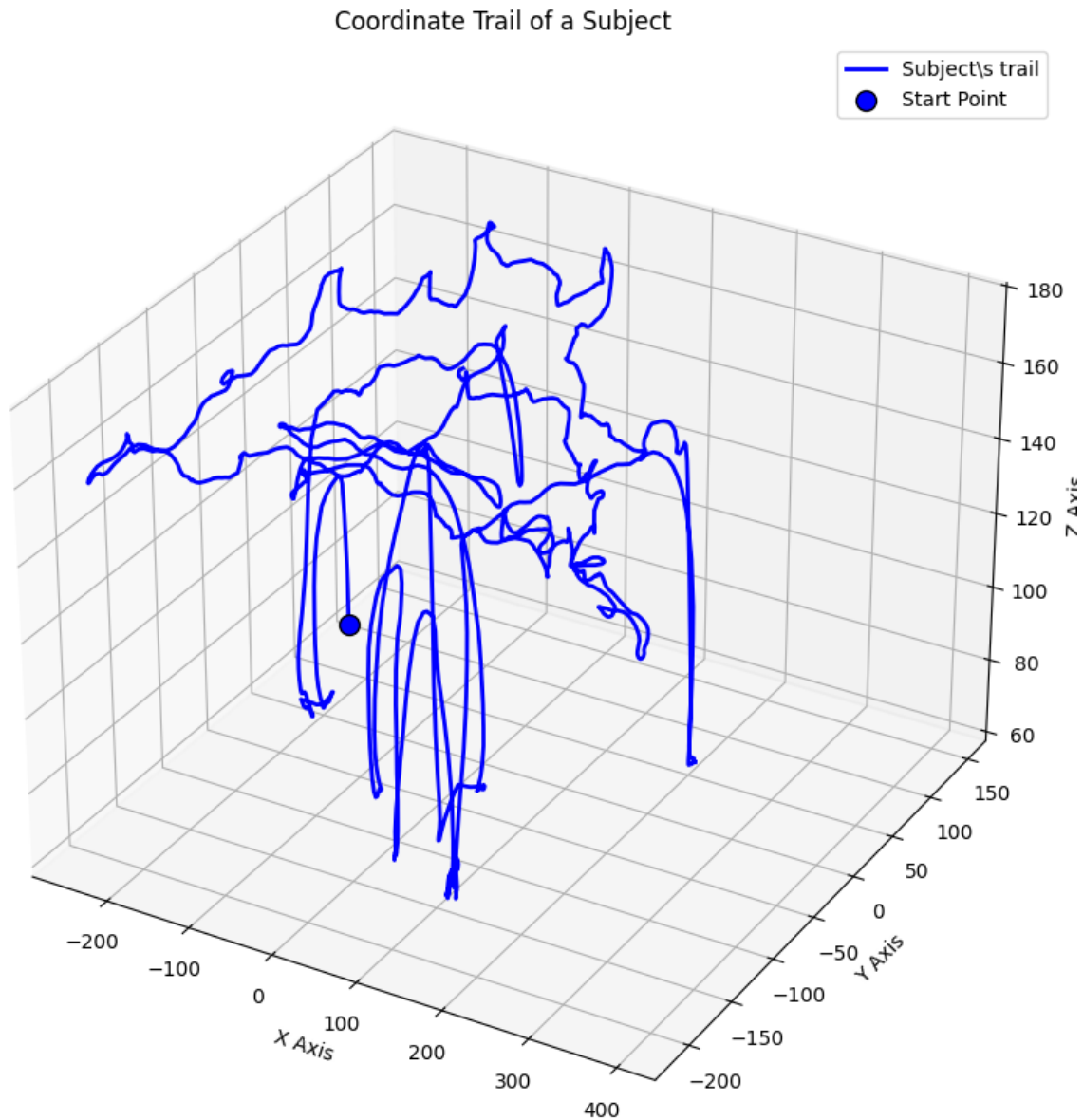


Figure 2. A visual trajectory of a subject's trail with the axis measured by cm.

The basket's location which initialized data collection was not randomized, and as such, all subjects started in a similar location. However, due to differences in height and posture, the exact starting position changed slightly. These variations could not be easily removed without causing serious disorientation to the participant or skewing the position of the environment for

many complexity tests, making the data unusable. While limiting a few potential complexity scores, a vast majority were unaffected.

Out of the 108 subjects recording, four were removed. Subject 1's data was unfortunately useless, as the Hz was incorrect for their experiment, resulting in only one coordinate being stored per second. Subject 67, 68, and 69's data were unable to be retrieved. An unknown bug in the Quest 3 resulted in all three logs being deleted before they could be retrieved. All other subject data was able to be collected and was error-free.

3.2.2 Feature Extraction

Given the high dimensionality of raw 10 Hz trajectory data, and the limited sample size of 104 subjects, a direct comparison of raw coordinate data was statistically unfeasible. To translate the high dimensionality to comparable metrics, the large amount of 3D data was converted into individual scores in a single dimension, allowing the complexity metrics to be calculable. In contrast, a single aggregate complexity score would result in significant information loss.

Due to the novelty of matching movement complexity to creativity on an individual scale, there was a notable absence of a pre-established gold standard. As a result, a discovery-oriented approach was selected, ranging from theoretically informed measures of information theory like Shannon Entropy, to exploratory kinematic markers. This exploratory approach allows for the identification of specific behavioral signatures that may correlate with creativity.

To evaluate these trajectories, 18 metrics were identified and categorized into four primary dimensions of movement:

- **Kinematic and Directional Complexity:** This included metrics such as Exploration, Coverage, Turns, Reverse Turns, Local Tortuosity, and Jitter.
- **Heatmap Analysis:** Heatmaps were created from each subject's data, allowing metrics such as Cold Cells, Hot Cells, Ratio, and Concentration
- **Displacement Information:** A collection of Mean Standard Displacement measures, such as MSD Slope, Frequency, Distance, Tempo, Consistency, and RMSD, were used to analyze how participants move relative to their starting point over time.
- **Entropic Measures:** Shannon Entropy and Entropy Rate were used to measure the uncertainty of the movement trails.

The first method of analysis, Exploration, focused on the subject's interaction with the physical constraints of the virtual forest. The physical extremes beyond the boundaries were recorded to quantify the breadth of their search.

Coverage was measured by first getting the spatial data of the subject with identifying the maximum boundaries reached by any subject across 2D, discounting the Z axis, and 3D space. With a spatial range of a meter around the subject's head, the total coverage of the space that the subject moved through was calculated and returned as a percent of the total space covered. The difference of percentages was measured between 3D and 2D space for the subjects, and the differences between them were identified to be consistently too small to make a difference, so 2D space was selected for ease of computation.

To quantify the amount of times the subjects turned, Turns and Reverse Turns were calculated using a sliding window vector analysis. For each data point starting at 16 due to a window size of 15 (1.5 seconds), two vectors were calculated. One vector recording the previous 1.5 seconds, while the second recording the next 1.5 seconds. With these vectors, the angular change was determined using the dot product of the vectors and their magnitudes:

$$\cos(\theta) = \frac{\text{dot}(v_1, v_2)}{\|v_1\| \|v_2\|} \quad (1)$$

The resulting cosine value was then converted to degrees to categorize the movement. Anything above 90 degrees represented a turn while anything exceeding 150 degrees represented doubling back, or a 'Reverse Turn'.

To ensure that a single turn was not double counting, a 20 point (2 second) cooldown was implemented. Once a turn was registered at the center of a window, the algorithm skipped forward 2 seconds before searching again.

Local Tortuosity served a similar function, but via a more abstract method, measuring the "winding" nature of the path rather than specific angular events. The movement path was segmented into windows of 50 data points (5 seconds) with a 50% overlap to ensure smoothness. For each window, a Tortuosity Index was calculated as the ratio of the actual path length to the straight-line displacement:

$$T = \frac{\sum_{i=1}^{n-1} \|P_{i+1} - P_i\|}{\|P_n - P_1\|} \quad (2)$$

Where the numerator represents the sum of all Euclidean distances between consecutive points within the window, and the denominator represents the net distance between the window's start and end points. A result of 1.0 indicated a perfectly linear path, while higher values quantify the degree of deviation or "wandering" within that window. This provided a measure for the characterization of the subject's movement efficiency that turn counts might overlook, such as long gradual curves.

In contrast to the other macro-level kinematic measures, Jitter quantified the constant micro-adjustments made by subjects. Another sliding window of 10 data points (1 second) on each side of the central point was used to create a smoothed trajectory. For each central point, a mean position was calculated using the points within the window to filter out high frequency noise. The Jitter score was then defined by the distance between the raw coordinates and the smoothed trajectory.

The metrics in Displacement Information are dependent on Mean Standard Displacement (MSD), which is a measure of the squared distance the subject moved from their starting point. By itself, this measure was not useful due to the limited nature of the space but provided a solid basis to build upon. When compared to time, it created a graph from which many different measures could be taken.

A threshold of 0.5 meters from the initial starting coordinate was established to define a 'return event'. From these events, the following five dimensions of displacement were extracted:

- **MSD Slope:** This metric represented the rate of displacement over time. A higher slope indicated a subject who consistently moved away from their starting point, while a lower slope indicated a more concentrated search around the origin.
- **MSD Frequency:** This metric recorded the raw count of return events.
- **MSD Tempo:** This measured the average peak displacement reached before a subject doubles back to return, providing a scale of each foraging excursion.
- **MSD Distance:** This measured the average peak displacement reached before a subject doubled back to the origin, providing a scale of each foraging excursion.
- **MSD Consistency:** To distinguish between rhythmic and stochastic navigation, MSD Consistency was calculated as the standard deviation of the return intervals. A low

standard deviation indicated a highly periodic, rhythmic search pattern, whereas a high value suggested a more erratic or non-linear approach to the environment.

- **RMSD:** Root Mean Squared Displacement computes the average distance away from the start point across the entire path.

The first measure within the Entropic measures was Shannon Entropy. Rather than measuring the linear sequence, this metric determined the stochasticity of the pattern, identifying subjects whose pattern was predictable compared to those whose pattern was more ‘uncertain’. To measure Entropy Rate, Markov trajectory entropy was utilized, which calculates the probabilities of different states of the trajectory, dependent on the previous state. This results in a calculation which quantifies sequence dependent unpredictability of the trajectory. This can be used to create a graph, from which the slope was used as a metric. However, Shannon entropy does not use temporal data, focusing only on positional distribution. To measure entropy over temporal data, measures such as Markov trajectory entropy incorporate the probabilities between different states, enabling an analysis of sequence-dependent entropy.

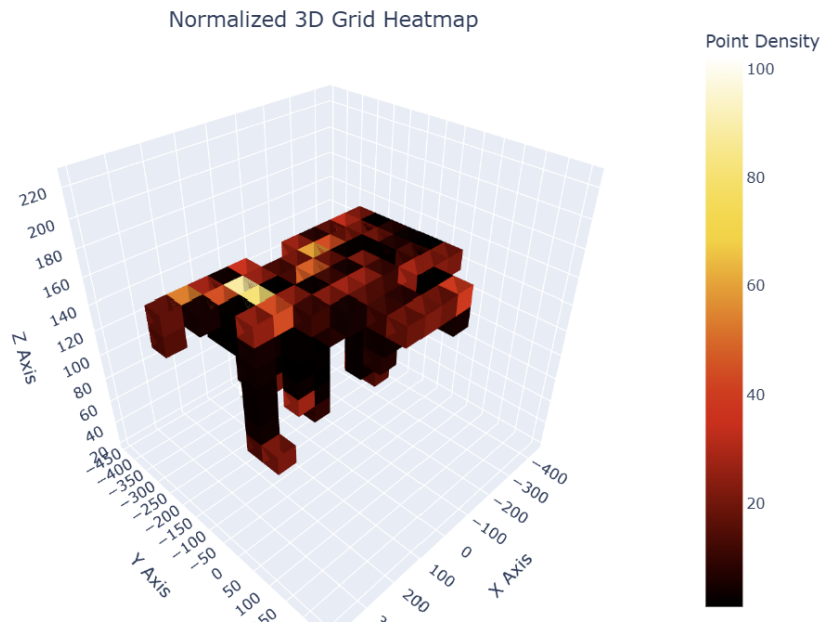


Figure 3. A normalized heatmap of a trajectory.

For the rest of the movement measures, Heatmap Analysis, a heatmap was created from the movement trajectories as can be seen in Figure 3. While grid sizes with fixed units were initially tested with a range of sizes, this approach was identified as overly sensitive to variances in the subject’s maximum ranges and height, creating noise within the data. As a result, a relative grid was adopted with a consistent number of bins across the axes (15x15x15). This normalized

approach isolated the structural complexity and thoroughness of the search itself, ensuring that the behavioral strategy was measured instead of the physical dimensions of movement which were covered by other tests.

The Cold Cells metric was determined by the amount of cells that had less than a 15 datapoints within them, yet more than 0. This indicated how many cells that the subject visited, but spent very little time in. Conversely, Hot Cells counted how many cells contained 45 datapoints or more, indicating the number of places the subject spent a lot of time in. Ratio is the ratio of the subject's hot cells to their cells. To create the Concentration metric, the heatmap was treated as a probability distribution. After flattening the 3D grid into a 1D vector, the cell counts were normalized by total residence time to calculate the probability of the subject occupying any grid cell. Shannon Entropy was then applied to this distribution to quantify the predictability of the subject's search. In this case, lower entropy indicated a higher concentration in the search pattern, whereas a high score reflected a more diffuse distribution.

3.3 Statistical Analysis of Confounds and Comparisons

To ensure the validity of the relationship between movement complexity and creativity, several potential confounding factors were measured and evaluated, which could be separated into three categories.

The first category was demographic information, which included Age, Sex, Education, and Occupation. Age was included as it had potential to alter movement patterns in various ways, most expectedly with subtle movement patterns such as Jitter. Sex was included to account for potential gender-based differences in the movement or creativity tests. Education measured the level of education the participants had, ranging from a 1 with basic education to a 7 for a doctorate. This was included to identify whether levels of education altered how people thought and moved. Occupation required minor encoding. If the subject was neither studying nor working, they received a 0. Studying received a 1, working received a 2, and a 3 if the subject was both working and studying. This provided a good way to see if cognitive load is affecting their movement.

The second category contained pre-existing experience. Given that the participants were searching for berries in a virtual forest, Foraging Experience was included to see if foraging in a VR environment was affected by foraging experience in the physical world. VR Experience was highly anticipated to have an affect on the data, as this prior could have significant effects.

People unused to the headset might not move their head different or worry it would fall off, discomfort could lead to distraction or reduced focus, and a difference in confidence could alter their behavior.

Additionally, two binary measure were recorded: Creativity Test Experience and Estonian First Language. Creativity Test Experience was a simple yes-no question on whether they'd performed in other creativity tests or experiments, which could predispose them to certain behaviors. Estonian First Language was deemed highly important as several of the tests are highly language-dependent and were conducted in Estonian.

The last group measured factors that could affect the energy of the participants. The first of these was Time, which measured the time of day, which was then turned into a numeric form for calculations. Tiredness was measured on a scale of 1 to 5 with 1 indicating 'not tired at all' and 5 indicating 'very much. Mood was also measured from 1 to 5, with 1 meaning 'bad' and 5 meaning 'good.'

Building upon the discussed constraints of the high dimensionality, the analytical strategy necessitated a controlled statistical framework. With a sample size of 104 subjects, the use of complex Machine Learning architectures, such as Neural Networks, would result in significant overfitting and unreliability. To analyze the data precisely, Spearman Correlations, Distance Correlations, Partial Spearman Correlations, and Partial Distance Correlations were selected as the primary analytical tools. These provided the ability to isolate the relationships between movement and creativity scores, independent of confounding factors such as 'sex' or 'VR Experience'.

To begin, correlation analyses were performed between every creativity score or potential confounds with each movement score. When comparing Spearman to Pearson correlations, Spearman was chosen as Pearson correlation is highly sensitive to non-parametric data. A number of the various scores were non-parametric such as Local Tortuosity and Jitter, as can be seen by the histograms in Figure 4.

With the correlations created between every comparison of data, a heatmap was created to indicate the strength of the correlations with three levels of statistical significance measured at 0.05, 0.01, and 0.001. While useful for seeing how complexity scores might be influenced by some confounding factors, the amount of potential noise that is in the data due to these confounding factors made measuring the correlation to creativity scores unstable.

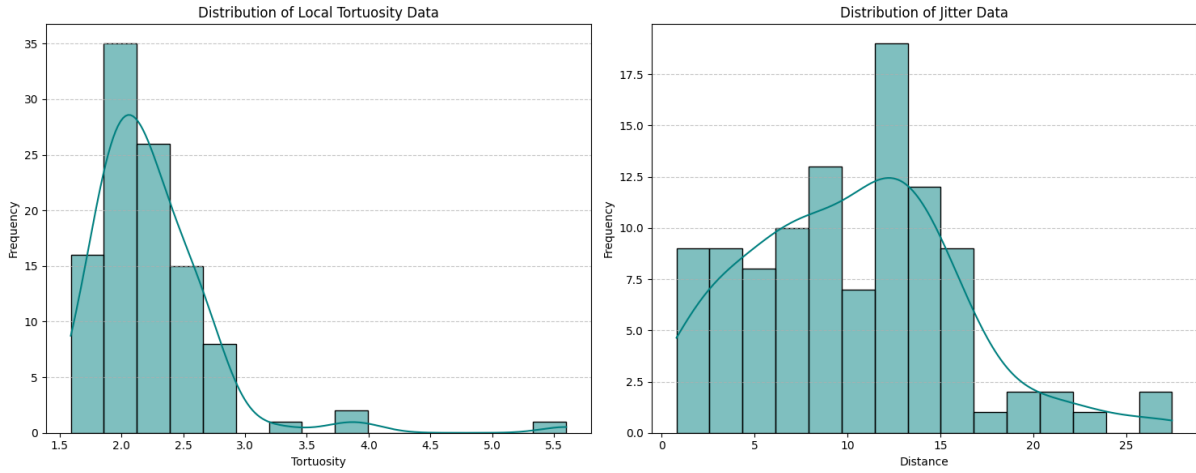


Figure 4. Non-parametric Data of Jitter and Local Tortuosity

However, Spearman correlation is sensitive to monotonic relationships and may fail to detect more complex non-monotonic associations. In contrast, Distance Correlations are specifically made to calculate correlations with non-linear and non-monotonic data, but does not check for positive or negative correlation. In response, it is common practice to create a sign-corrected distance correlation using the signs of Pearson or Spearman correlations. Given the other correlations use Spearman, this was once again used for the distance correlation signs.

For each pair of variables, the movement metrics were randomly shuffled 1,000 times while keeping the creativity scores fixed. The correlation between the shuffled movement metric and the creativity score was recalculated for every permutation. Because each permutation destroys the inherent relationship between the variables, this process creates a null distribution representing the correlations that would occur by pure chance. The number of 'fake' (permuted) correlations that exceeded the observed 'true' correlation was then used to calculate the p-value. To ensure a conservative estimate and prevent the occurrence of a p-value of zero, a standard correction was applied where 1 was added to both the count and the total number of permutations:

Distance Correlations do not have a simple analytic p-value. Therefore, a non-parametric permutation test was used to determine statistical significance. This process involved iterative sampling to build a custom distribution for each variable pair.

For each pair of variables, the movement metrics were randomly shuffled 1,000 times while keeping creativity scores fixed. The correlation between the shuffled movement metric and creativity is calculated every permutation. Because each permutation destroys any relationship between the variables, this process creates a null distribution even if the care correlation is strong

via random chance. The number of fake correlations which exceeded than the true correlation is then used to calculate the p-value. To prevent a p-value of 0 occurring, a small correction where 1 was added to both the count and total number of permutations:

$$P = \frac{\text{count}(\text{corr}_{\text{perm}} \geq \text{corr}_{\text{true}}) + 1}{n_{\text{perm}} + 1} \quad (3)$$

To isolate the relationship between movement complexity and creativity from the confounding factors, Partial Spearman Correlations were calculated using a residualization technique. This process ensures that the correlations are not artifacts of common variance shared with confounding variables.

For each movement metric and creativity score, a linear regression model was fitted using the control variables as predictors. By subtracting the predicted values from the observed data, we obtained residuals, representations of the portion of movement and creativity that remain unexplained by the confounds. A Spearman Correlation was then performed on these residuals. This approach was chosen to account for potential non-linearities in the movement data while maintaining the rigorous controls of a partial correlation framework.

Statistical significance was then determined using adjusted degrees of freedom ($df = n - 2 - k$), where n is the sample size and k is the number of controlled confounds. This adjustment is required to ensure that to prevent Type I errors with the p-values.

The distance correlations were then run on the cleaned data used from Partial Spearman Correlations. Because permutation testing is empirical, data-driven, and independent of distributional assumptions or analytic p-value formulas, no shift in p-value calculations were necessary.

Because a large number of movement complexity metrics were calculated, correlations between trajectory features were additionally analyzed to evaluate redundancy between measures. Spearman correlation matrices were used to examine relationships between movement metrics prior to dimensionality reduction. This analysis was performed to assess whether metrics captured overlapping or distinct aspects of exploratory behavior before the use of PCA and clustering for identifying broader latent structures.

To further explore the potential for dimensionality reduction, a Principal Component Analysis (PCA) was performed on the 18 movement metrics. To ensure a thorough investigation of the latent behavioral structures, PCA was tested with varying numbers of components included in

the model. Each resulting PCAs were then subjected to both Spearman and Partial Correlation analyses to determine if a collapsed dimensionality provided a stronger or more stable signal for creativity than the individual metrics.

To investigate whether movement trajectories formed naturally separable behavioral groups, k-means clustering was applied to the PCA-reduced feature space. The elbow method was used to examine reductions in variance within clusters across multiple cluster counts, while silhouette scores were used to evaluate cluster separation and cohesion. PCA component projections were additionally visualized to examine the spatial distribution and overlap between clusters within reduced dimensional feature spaces.

4. Results

4.1 Relationships Between Creativity Measures and Movement Metrics

By combining these non-parametric correlation techniques with a rigorous residualization framework, the analysis successfully isolated the behavioral signals of movement from the noise of individual demographic and experiential differences. The following subsection presents the findings of these analyses, beginning with the raw correlation landscapes before moving into the refined partial correlations and finally the structural insights provided by the dimensionality reduction and clustering.

The first correlations were run by comparing movement scores with creativity scores. The AUT and DAT scores are depicted on the left graph in Figure 5, while the self-reported scores are depicted on the right.

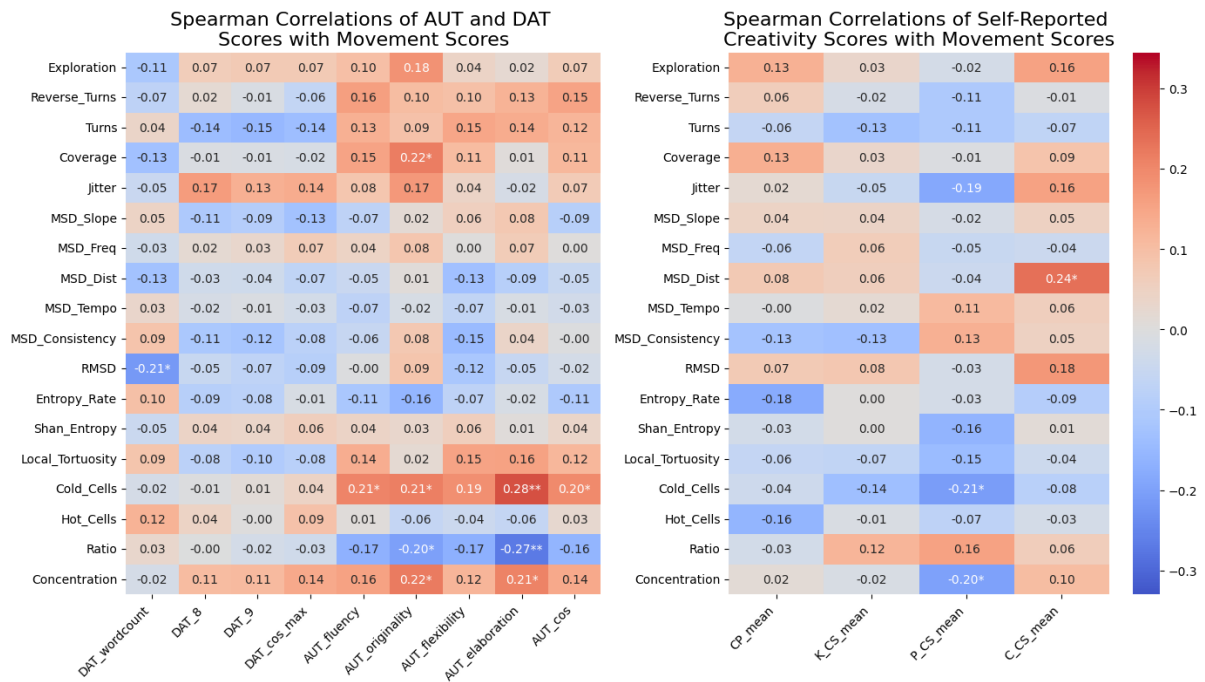


Figure 5. Creativity Scores' Spearman Correlations with Movement Scores. On the left are the AUT and DAT scores. On the right are the self-reported scores.

As can be seen, most correlations are small, but there are consistent patterns that can be noticed. AUT originality correlates the most often, being tied to Exploration, Coverage, Cold Cells, and Concentration with the highest being correlations with a coefficient of 0.218484 and a p-value of 0.0259. However, the creativity metric that correlated the strongest was AUT Elaboration with a correlation strength of 0.277695 and p-value of 0.0043 with Cold Cells.

Among the self-reported creativity scores, the structure differs significantly. There are significantly few results, one of which stood out unusually. The Creating domain of thinking correlated with MSD Tempo, whereas the only other creativity measure to correlate with any MSD scores was DAT wordcount.

Notably, certain movement scores held patterns of their own. Cold Cells consistently correlated much more-so than any other metric, followed by Ratio which is dependent on Cold Cells. Additionally, several movement metrics consistently correlated inversely across both types of creativity scores, such as RMSD, Ratio, and Entropy Rate.

Additionally, the correlations between movement metrics and potential confounding factors were calculated and displayed in Figure 6. The confounding effects are strong, with correlations being both stronger and more plentiful. Turns contained two of the three strongest correlations, with the strongest being tied to VR Experience with a correlation coefficient of 0.344674 and a p-value of 0.000972.

Several factors contained strong correlations across many movement metrics, with Foraging Experience, VR Experience, and Sex containing the strongest and most consistent correlations. While indicative of significant noise within the dataset, the awareness provided has significant value.

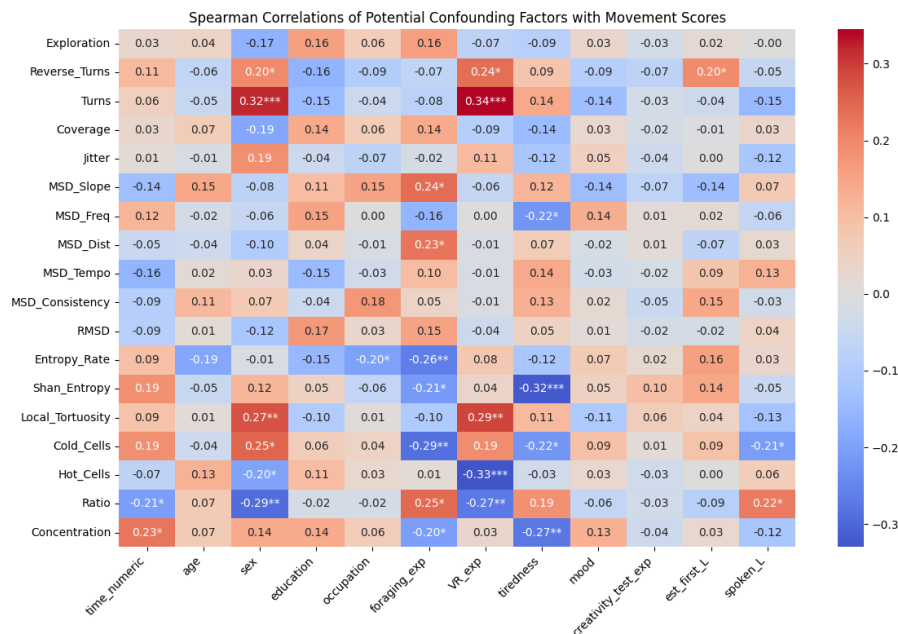


Figure 6. Potential Confounding Factors' Correlations with Movement Metrics

While Partial Spearman Correlations account for confounding variables, they remain limited to monotonic relationships. To capture more complex associations and verify that correlations were not missed, Distance Correlation analyses were conducted on both the creativity scores and the potential confounding factors as seen in Figure 7

The results of the Distance Correlations identified a similar structure to the Spearman Correlations, verifying their consistency. Additionally, Distance Correlations revealed higher correlations, which were to be expected from the higher sensitivity of the method. The requirement for statistical significance was higher as a result, but still revealed several new associations.

AUT correlations with Cold Cells, Concentration, Coverage, and Ratio and remained higher than most other correlations, but the stricter requirements core statistical significance resulted in less relationships. AUT Elaboration in particular had much stronger correlations overall, including the strongest correlation of 0.318048, which was an inverse correlation.

Several new correlations appeared as well, with Exploration and Reverse Turns correlating with two different scores each from the AUT or the Creating Domain. Self-reporting scores matched the previous structure as well, with the most notable difference being that the inverse correlation between the Planning cognitive domain and Cold Cells disappeared.

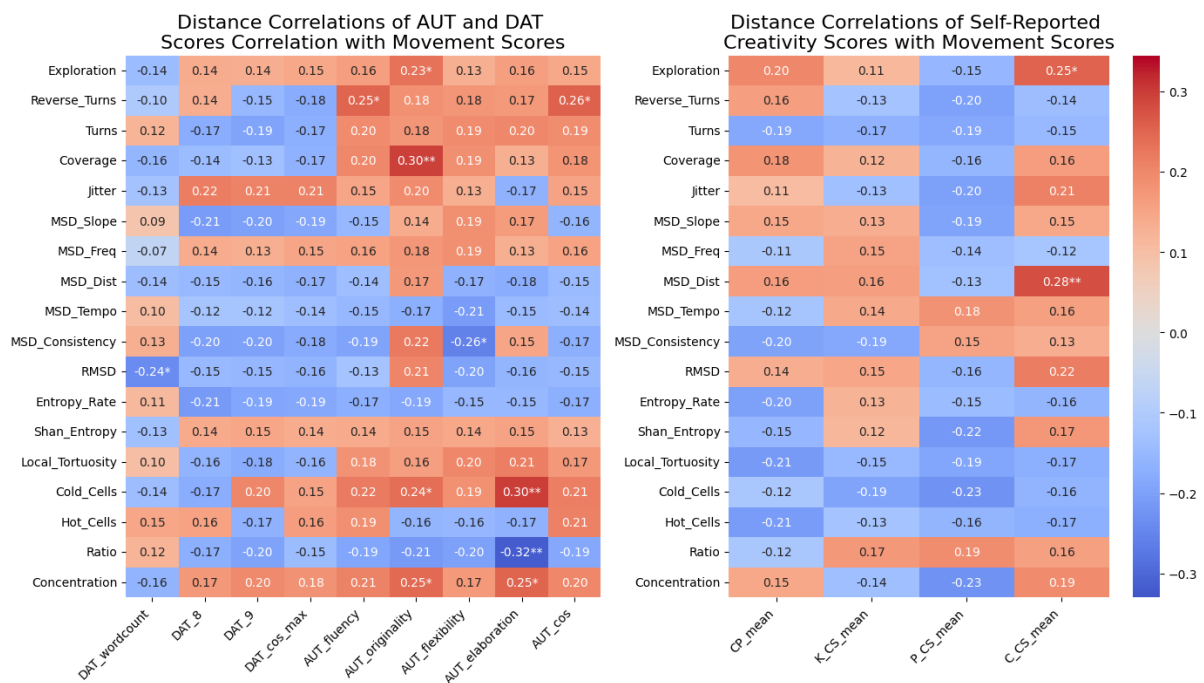


Figure 7. Creativity Scores Distance Correlations with Movement Scores. On the left are the AUT and DAT scores. On the right are the self-reported scores.

Despite revealing additional non-monotonic associations, these results remained influenced by confounding variables similar to the Spearman Correlations. The strong correlations with confounding variables for Distance Correlations matched nearly exactly, with only minor variances of 0.1 across them with one exception. The correlation between VR Experience and Cold Cells dropped significantly, to a point where it was no longer statistically significant.

4.2 Creativity Relationships After Controlling for Confounds

After controlling for confounding variables with residualizations, both Spearman Correlations and Distance Correlation were recalculated. For the Spearman Correlations, most weak correlations attenuated, while the structure of the strong correlations intensified as shown in Figure 8

Metrics reflecting spatial exploration and distribution, particularly Cold Cells, showed consistent positive correlations with AUT measures with one exception. AUT elaboration dropped to 0.173904, and was no longer statistically significant. The correlation with AUT Originality and Coverage dropped slightly, but remained strong enough to be statistically significant. Lastly, one new correlation appeared with Reverse Turns correlated with AUT cos.

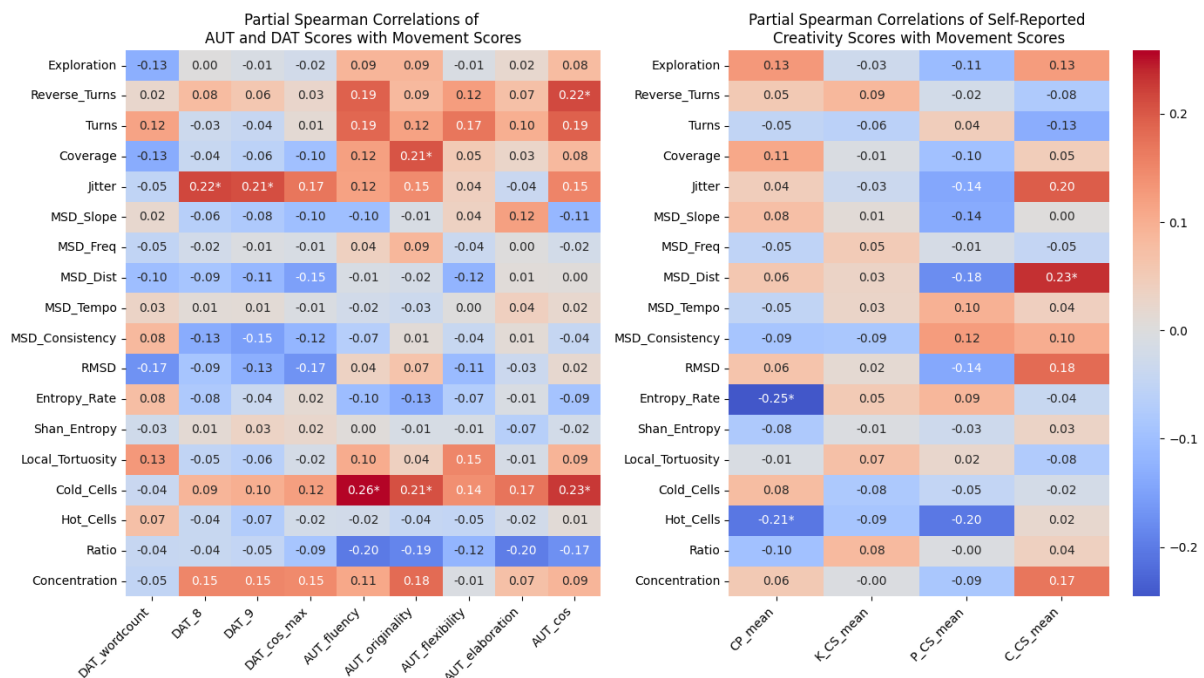


Figure 8. Creativity Scores Partial Spearman Correlations with Movement Scores. On the left are the AUT and DAT scores. On the right are the self-reported scores.

In tandem, Jitter rose from weak scores to being statistically significant across multiple DAT metrics. This measure indicates micro-movements in contrast to the general exploration and macro movements that coverage and Cold Cells do.

Similarly, the self-reported scores showed significant increase in correlation strengths in several areas. CP Mean, the self-reported creativity, increased to significant correlations with both Hot Cells and Entropy Rate, while the Creating domain remained at a similar level.

In contrast to the Spearman Correlations polarizing, the strong Distance Correlations diminished slightly while the weak correlations became unstable.

Many of the weak correlations near 0.15 switched between positive or negative, but this was dependent on the Spearman Partial Correlation. Given that Distance Correlations does not have a sign attached to it, the positive or negative direction from the Spearman Correlations were used. Because Spearman Correlations were close to 0, tiny fluctuations could flip the sign, resulting in the unstable relationships seen within Distance Correlations. As such, this doesn't indicate a real reversal or relationship, merely that no stable directional relationship exists.

In contrast, many of the strong results remained stable. Even after controlling for confounds and analyzing for non-monotonic relationships, the same correlations remain consistent. AUT Fluency and AUT Elaboration remain correlation to Cold Cells, and AUT Originality contained the second highest correlation with a strength of 0.259853 and a p-value of 0.018981. The only higher correlation was the Creating cognitive domain with MSD Distance, which has remained consistent through all tests. The self-reported creativity tests maintained a consistent structure as well, with Planning correlating to Entropy Rate. While this didn't appear in the Distance Correlation when not controlling for variables, this is consistent with the Partial Spearman Correlation.

As an additional robustness check, two cognitive style measures, Knowing as K_CS_mean and Planning as P_CS_mean, were separated from the other creativity scores due to theory suggesting their lack of relation with creativity. To ensure that they were not acting as confounding variables, a separate partial correlations heatmap was recomputed with the two scores added to the confounding variables. The inclusion of these variables resulted in minor attenuation of correlations involving AUT and DAT scores, while slightly strengthening associations with self-reported creativity measures. However, these changes were small, generally around 0.1, and no major patterns of significance were altered.

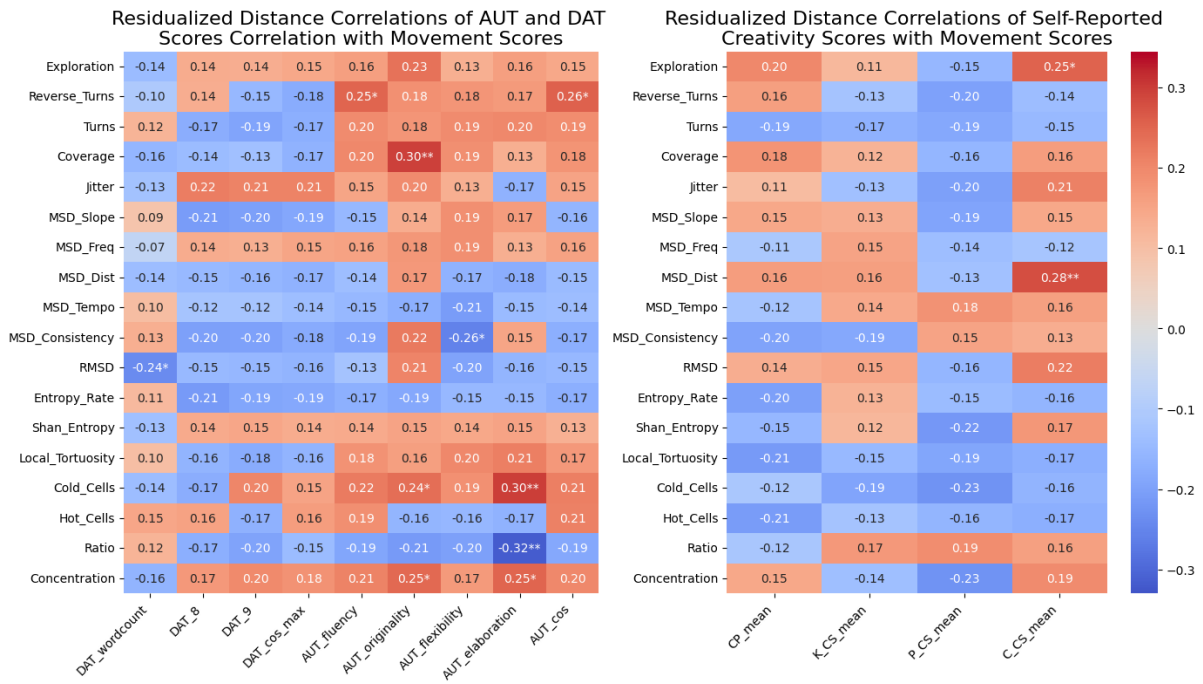


Figure 9. Creativity Scores Partial Distance Correlations with Movement Scores. On the left are the AUT and DAT scores. On the right are the self-reported scores.

4.3 Internal Structure of Movement Complexity Features

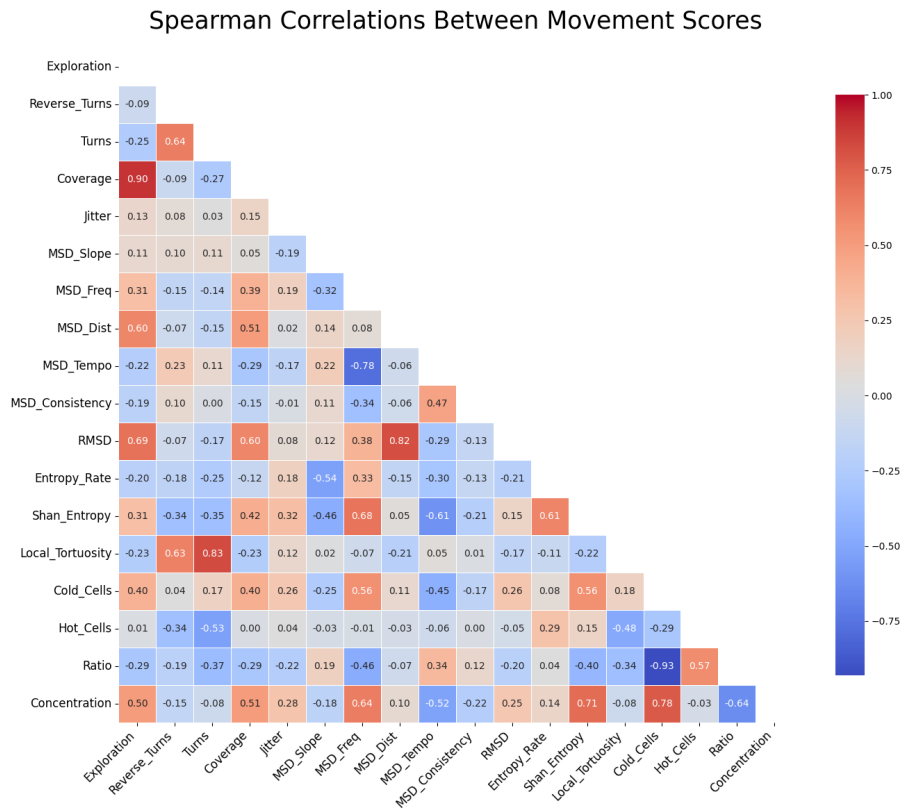


Figure 10. Correlations between Movement Complexity Metrics

To identify potential redundancy and interdependence between complexity metrics, correlations between movement features were analyzed. Several metrics displayed strong correlations, particularly between related measures such as Exploration and Coverage, RMSD, and MSD Distance, and Turns and Local Tortuosity. The strongest relationship was an expected inverse relationship between Cold Cells and Ratio.

These results indicate that while some redundancy exists between movement features, no single metric dominated the space. Due to the high dimensionality and partial interdependence of the trajectory measures, dimensionality reduction techniques such as Principal Component Analysis (PCA) were used to identify broader latent structures within the movement data. Spearman Correlations were then performed, allowing the correlations to be compared to the internal insights of PCA.

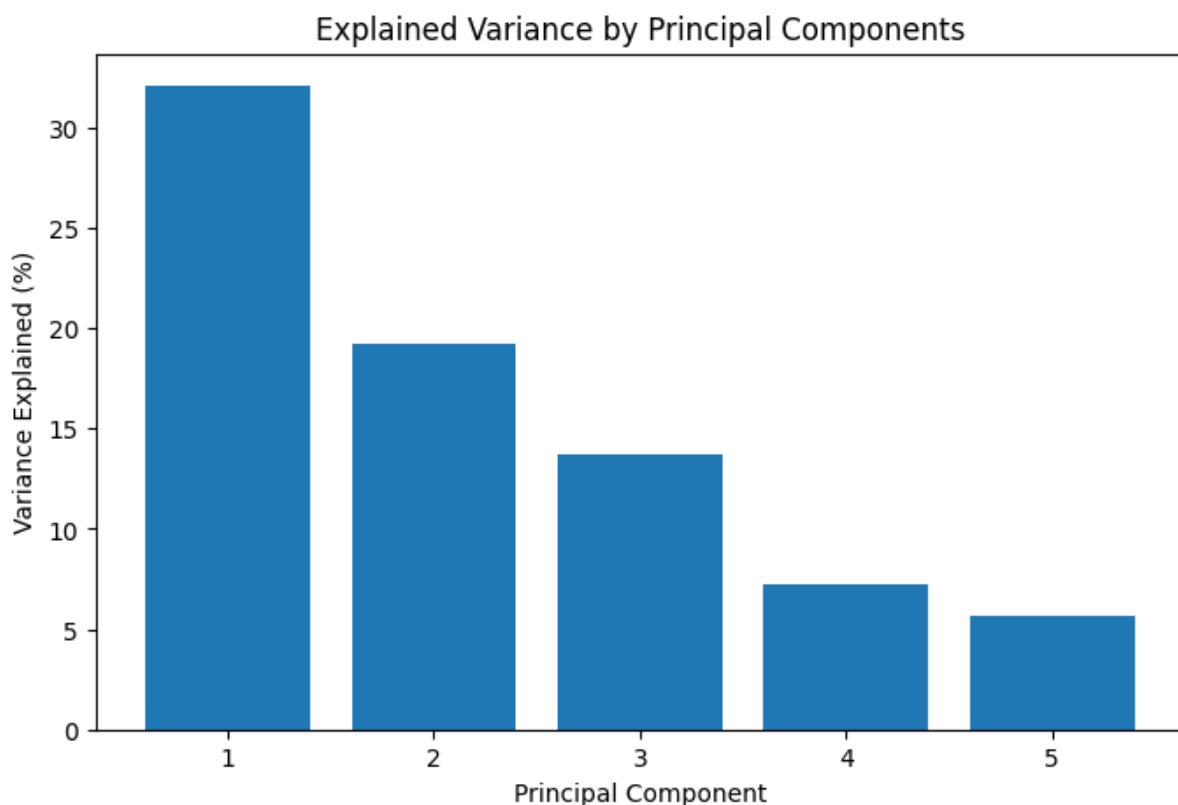


Figure 11. The explained variance by each component of the PCA

The first PC explained 32.4% of the variance and the second PC explained 19.5% of the variance. With the other three PCs combined explaining 25.1% of the variance, the total variance was 77%, showing that a large proportion of the variance was captured within the first few components.

The loadings of the first PC show positives which are focused on heavily exploratory measures that stay away from the origin point such as Cold Cells, Coverage, and RMSD. Conversely, the strong negatives focus on repetitive behavior such as MSD Tempo and MSD Consistency.

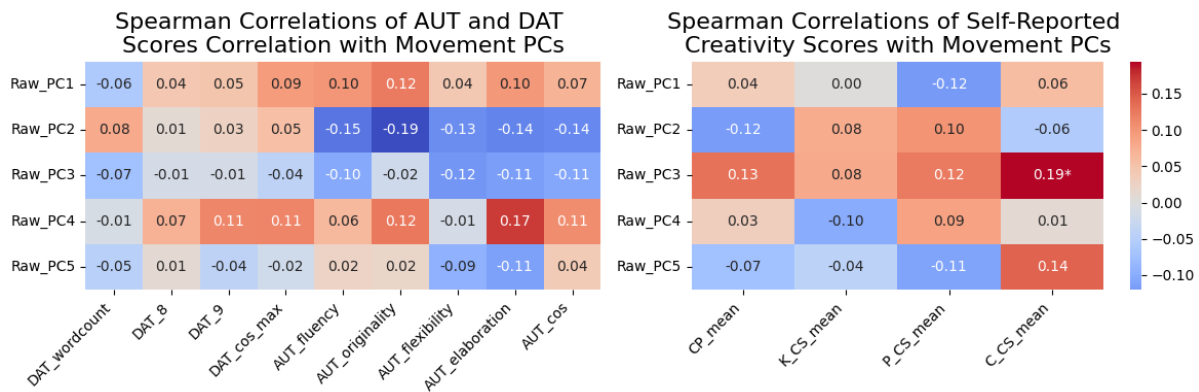


Figure 12. Creativity Scores Spearman Correlations with PCs. On the left are the AUT and DAT scores. On the right are the self-reported scores.

These patterns suggest that PC1 was heavily focused on diffuse exploratory movement, the same theory and measures supported elsewhere in this research and other research. However, PC1 is not strongly correlated with any creativity measure. This suggests that that creativity relationships are likely tied to more specific movement features as some loadings could have reduced the correlation such as Shannon Entropy, which was a strong loading, but has been uncorrelated or inversely correlated in other tests. In contrast, it ties closely with tiredness, which aligns with the chaotic nature that this PC focuses on.

PC2 had fewer positive loadings, but they had much higher strengths, consisting only of Hot Cells, Ratio, Entropy Rate, and Shannon Entropy, nearly all of which showed variations of inverse correlations. The strongest negative loadings included Cold Cells, Reverse Turns, Local Tortuosity, and Turns, many of which contained strong correlations. As can be seen in Figure 12, this strong implication of PC2 inversely correlating with creativity is consistent with the observed correlation patterns, as while never statistically significant, PC2 showed many weak inverse correlations. However, this PC also appears to be strongly correlated with VR Experience, indicating potential noise.

With a variance of 12%, PC3 contained the only statistically significant correlation with the Creating cognitive. Within the positive loadings, PC3 focused heavily on MSD, containing 3 of the top 4 loadings, all of which focused on how far the subject moved away from the origin. Meanwhile, the negative loadings focused on Entropy Rate, Turns, MSD Frequency,

and Shannon Entropy. This indicates that this PC is opposed to frequent and local movement. This contained the strongest correlation out of the PCAs with Foraging Experience, as well as negative with VR Experience.

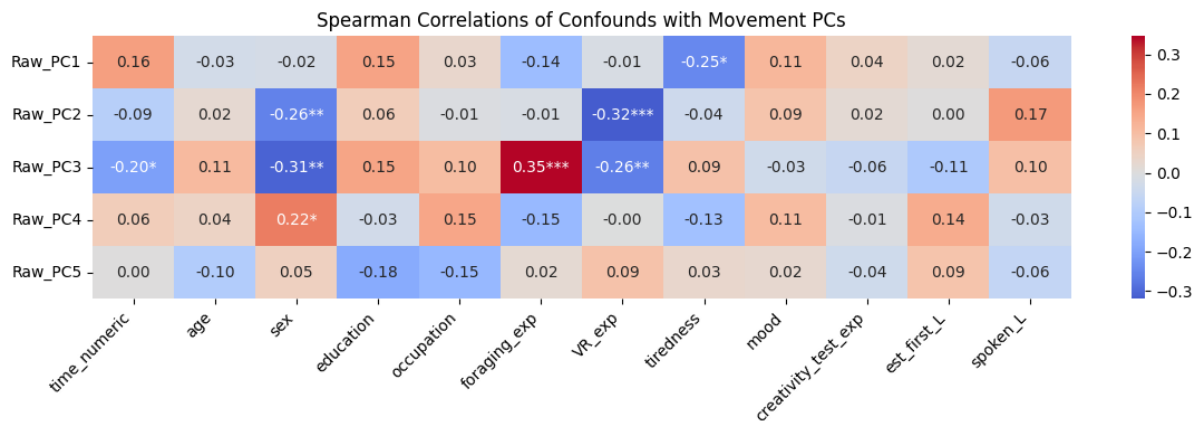


Figure 13. Potential Confounding Factors' Correlations with PCs

Because PCA was performed on standardized variables using a StandardScaler, each feature contributed unit variance to the analysis. As a result, the total variance explained by the principal components remained constant before and after residualization. However, residualization altered the correlation structure between variables, leading to differences in the resulting component loadings and their relationships with creativity measures.

The loadings for PC1 remained largely stable. The top positive and negative loadings all remained similar, while several central loadings changed. Similarly, the correlations remained quite similar between the two. With both remaining stable, this indicates the PC1 was largely independent confounding factors. PC2 also remained fairly stable. The relative position of loadings compared to each other was even more stable than PC1, but the values themselves changed. Positive loadings were reduced while negative loadings became stronger. The results can be seen in the correlations as many correlations also change, becoming stronger inverse correlations in most spaces except for AUT originality.

PC3, 4, and 5 all changed more dramatically within the loadings, and this can be seen in the correlation changes in PC3 and 4. However, the changes in PC5 had little effect on correlation strength. PC3's positive loadings changed to focus more on exploration and coverage, and the changes can be seen in correlations as it is no longer statistically significant. Meanwhile, PC4's change occurred in a similar manner to PC2, where a large amount of the change came from positive loadings decreasing. Several notable loadings changed from positive to negative,

including exploration and coverage. Given that the strong positive loadings were also focused on exploration, the positive correlation with the Planning cognitive style is a consistent result.

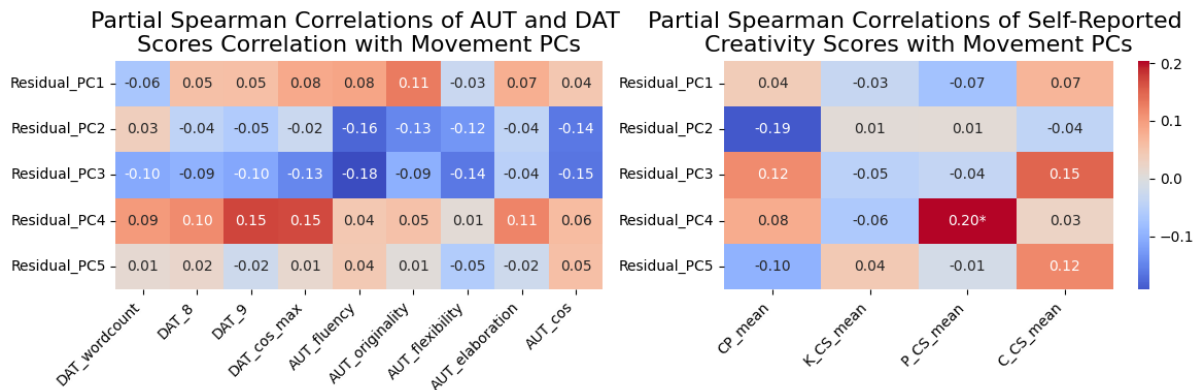


Figure 14. Creativity Scores Partial Spearman Correlations with PCs. On the left are the AUT and DAT scores. On the right are the self-reported scores.

To investigate whether exploratory movement behaviors naturally formed separable groups, k-means clustering was applied to the trajectory metrics and PCA-reduced feature space. Elbow and silhouette analyses indicated weak cluster structure overall, with only minor separation observed after dimensionality reduction.

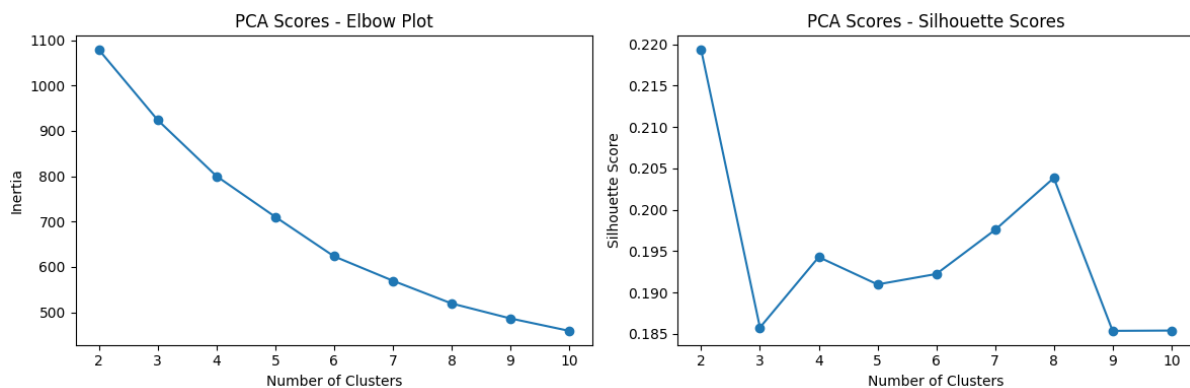


Figure 15. Elbow and Silhouette Scores for k-means clustering of PCAs

Visualization of the PCA features revealed two patterns across component projections. Some demonstrated substantial overlap between clusters, indicating little separability. Other projections exhibited partial separation along a single axis, behaving similarly to a linear gradient than distinct cluster boundaries. Together, these findings suggest that exploratory movement behavior does not form clearly separable behavioral categories.

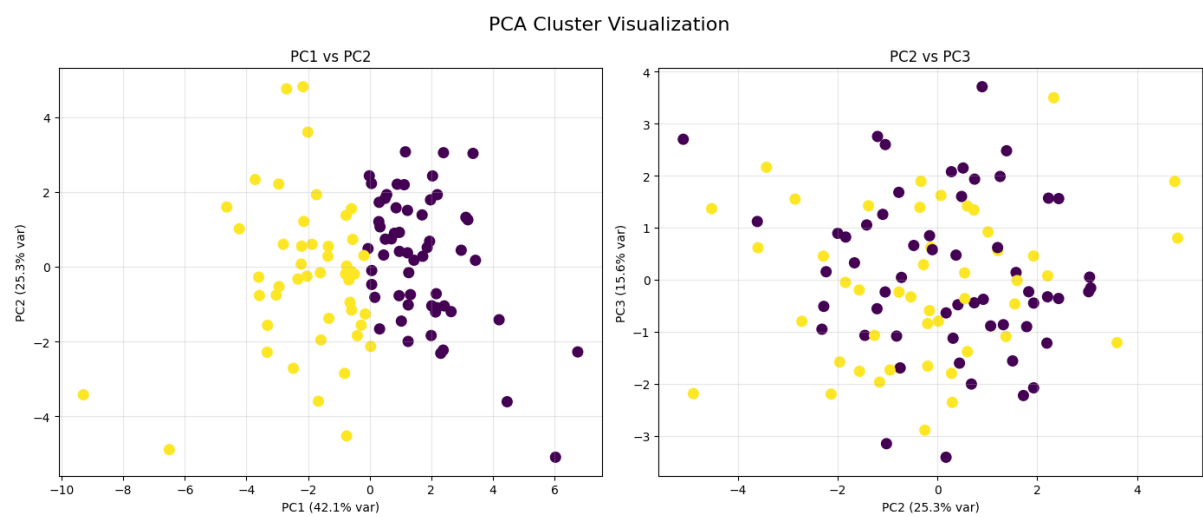


Figure 16. Visualization of k-means clustering for PCA

5. Discussion

5.1 Interpretation of Findings

This study investigated whether various dimensions of creativity were associated with exploratory movement behavior in a controlled VR environment. The findings provide insights into the relationships between movement complexity and creativity assessments with quantifiable metrics. Multiple different movement metrics were associated with different dimensions of creativity, supporting the multi-metric approach.

Creativity measures such as the DAT appeared to align more strongly with micro-adjustments, while AUT measures showed associations with broader exploration patterns. Consistent with previous theory, the results indicate that exploratory movement complexity is multidimensional, with different behavioral movement features associating with different creativity measures.

5.2 Limitations and Future Work

Several confounding factors exhibited stronger correlations with movement behavior than the creativity assessments themselves. In particular, prior VR and foraging experience demonstrated consistent relationships across multiple trajectory metrics, which could be controlled for in future experimentation.

Several participant related limitations could be eliminated in future research and experiments. To mitigate the effect of VR experience, expanded tutorials and improvements in VR hardware may reduce discomfort due to the headset or VR unfamiliarity. Additionally, a vast majority participants were working alongside studying, or only studying. This reveals a potentially biased sample pool across many demographics such as age and education, which could affect movement data or creativity measures.

The total size of the participant pool also supplies several issues. The dimensions of the data in relation to the sample pool limited the use of complex machine learning models. These findings use correlations for analysis and do not establish causal relationships. If an experiment were to collect enough data to perform such analytical methods, more relationships might be able to be identified.

Other additional limitations were identified which were a result of data tracking. With a framework already in place, more data measures could be utilized to expand on the experiment, notably hand and eye tracking systems. In testing, the correlation between the DAT scores and

Jitter only weakened as data was filtered out, suggesting that future systems utilizing higher frequency sampling and multiple methods of tracking may better capture behaviors associated with exploration.

This experiment was also limited by the environment. While the 4x6 meter area provided enough space to conduct the experiment, a larger area is expected to provide more variation in the data, and therefore stronger relationships with creativity metrics. Additionally, the activity of foraging for berries provided one of the strongest confounding factors, and a future study could try to limit prior experience by using a different activity.

Because the study was exploratory and aimed at identifying potential relationships across a broad set of behavioral metrics, emphasis was placed on sensitivity to potential associations rather than strict control of error rates. As a result, conservative multiple-comparison correction methods were not prioritized, as they could obscure weaker but informative behavioral associations in the exploratory analyses. As such, the reported findings should be interpreted as a foundation for future investigations into the relationship between exploratory movement behavior and creativity measures.

Overall, the findings support the possibility that exploratory movement behavior contains measurable signals related to creative cognition. While no single trajectory metric was sufficient to characterize creativity independently, the observed relationships across multiple metrics suggest that movement complexity may reflect different dimensions of exploratory cognitive processes. These results demonstrate the potential for future research to investigate exploratory movement as a tool for investigating creativity.

6. Conclusion

The primary objective of this research was to investigate the association between exploratory movement complexity and creativity measures of divergent thinking. To examine this relationship, participants performed a virtual foraging task within a VR environment. Movement trajectories were recorded, analyzed, and compared to multiple creativity assessments.

The results from the experiment reinforce that different movement metrics can be associated with different aspects of creativity. The Alternate Uses Test showed associations with broad exploration patterns while the Divergent Association Task showed associations with micro-adjustments. PCA and clustering analyses further reinforced that exploratory movement was distributed across multiple metrics.

Future research may further refine these findings by addressing several confounding factors and limitations affecting the work. Various confounding factors displayed stronger correlations than the individual creativity metrics, and the limited sample size prevented the use of complex machine learning approaches. Future work with larger datasets, improved VR systems, and additional behavioral tracking methods such as hand trajectories or eye movement may provide a more detailed understanding of the relationship between exploratory behavior and creativity.

In conclusion, the study demonstrates that behavioral patterns associated with creativity measures can be demonstrated through VR trajectory analysis. While additional refinement and validation are required, this framework supports continued research into movement analysis as a complementary approach for studying creativity and related cognitive processes.

7. Summary

Recent studies have suggested that creativity may reflect exploratory search processes within conceptual space. This can draw parallels between exploration in conceptual and physical space, suggesting that exploratory behavior may contain measurable signals for creativity. Other studies have similarly identified that movement complexity of hand movement has been linked with creativity, adding credence to the theory. This thesis investigated the association between exploratory movement behavior in virtual reality environments and creativity measures.

In this study, virtual reality was used to collect movement data while 104 different participants performed a searching task in a virtual forest. Multiple creativity assessments were used to capture various aspects of creativity, including the Alternate Uses Test (AUT), Divergent Association Task (DAT), and Beghetto's Creative Self-Efficacy. Movement trajectories were analyzed using various entropic, spatial, and kinematic measures. To explore potential relationships between movement complexity metrics and creativity measures, dimensionality reduction and comparative analyses were then performed.

With initial findings, very few relationships between creativity tests and movement complexity metrics were visible with only two AUT measures and the creating cognitive dimension correlating. Most creativity measures and potential confounding variables had weak correlations. However, several confounding variables appearing, consisting of stronger relationships than any creativity metric.

The data structure was then residualized to control for confounds, removing the effect of confounding factors on the data. The relationships seen in the AUT persisted and strengthened while several new behaviors emerged. Divergent Association Task scores aligned more strongly with localized micro-adjustments, while Alternative Uses Test measures showed stronger associations with broader exploratory movement patterns.

To identify potential redundancy and interdependence between complexity metrics, dimensionality reduction and clustering was performed. However, no single movement dimension was dominant and significant overlap appeared between clusters without any clear structure. The distributed nature of the data reinforces and supports both the pre-existing theory as well as the use of multi-metric approaches.

Several limitations affect the data and strength of the observed relationships. Multiple confounding variables, particularly VR and foraging experience demonstrated stronger and

more consistent correlations with subjects than the creativity measures themselves. In addition, the sample size limited the complexity of machine learning models that could be utilized. Several technical limitations also reduced the effectiveness of the data, such as limited space and a tracking frequency of 10 Hz, which was discovered to be too low to analyze micro-adjustments. However, several relationships between the movement data and creativity measures remained consistent across multiple analyses, suggesting that movement trajectories provide a useful foundation for investigating exploratory cognitive processes in future research.

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